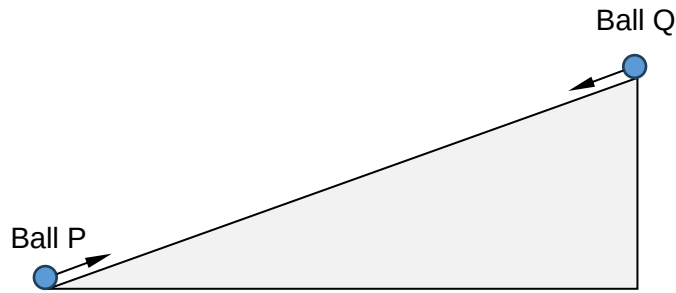


- 3 Ball P is projected upwards from the bottom of a smooth inclined plane with an initial speed that is just sufficient for it to reach the top of the plane.

At the same time that Ball P is projected, Ball Q is projected downwards from the top of the plane with the same initial speed as Ball P.



What is the ratio of the distance travelled by Ball 2 to the distance travelled by Ball 1 when they collide?

- A 0.78 B 1.0 C 1.3 D 3.0

Ans: C

From $s_1 = ut - \frac{1}{2}at^2$ and $s_2 = ut + \frac{1}{2}at^2$, where $a = g \sin \theta$

$$\Rightarrow s_1 + s_2 = 2ut \quad (1)$$

From $v^2 = u^2 - 2as$,

$$\Rightarrow s_1 + s_2 = \frac{u^2}{2a} \quad (2)$$

From (1) & (2),

$$\Rightarrow 2ut = \frac{u^2}{2a}$$

$$\Rightarrow t = \frac{u}{4a}$$

$$\frac{s_2}{s_1} = \frac{ut + \frac{1}{2}at^2}{ut - \frac{1}{2}at^2} = \frac{u + \frac{1}{2}a\left(\frac{u}{4a}\right)}{u - \frac{1}{2}a\left(\frac{u}{4a}\right)} = \frac{\frac{9}{8}}{\frac{7}{8}} = 1.3$$

- 4 The rate of change of momentum experienced by a free-falling ball when it hits the ground is